The Arithmetic Site and Persistent Homology: Prime Emergence as Dimensional Expansion over $\mathbb{F}_1$

Abstract

This monograph presents a novel topological framework for the emergence of prime numbers. Rather than viewing the integers ($\mathbb{Z}$) as an a priori one-dimensional continuum, we model $\mathbb{Z}$ as the emergent boundary where two distinct geometric realities intersect: the purely multiplicative lattice of the Arithmetic Site over the field with one element ($\mathbb{F}_1$), and the one-dimensional additive continuum. By constructing an Additive-Multiplicative Simplicial Complex and applying persistent homology, we demonstrate that prime numbers function as dimensional basis generators that resolve topological tensions. The persistent H_0 barcode encodes prime gaps, while H_1 cycles encode the braided computational complexity of factorization. We further define an Arithmetic Jones Polynomial to encode factorization paths, revealing that the topological cost of factorization is conserved within the $\mathbb{F}_1$ tangent space of its base prime.

1. Introduction: The Dual Generators of $\mathbb{Z}$

In standard arithmetic, the integers form a ring equipped with addition and multiplication. However, this structure obscures the profound asymmetry between the two operations. We propose a geometric re-interpretation based on the structural roles of the operational units:

1. **The Additive Identity (0):** The origin of the space.
2. **The Additive Unit (1):** A strict *translational* operator that shifts the space linearly.
3. **The Multiplicative Units (Primes p_i):** *Rotational* or *dimensional* operators that generate orthogonal lattice axes.

In this framework, the integers are not a line, but a braided topological fabric. The primes serve as the "warp" (structural basis generators forming an infinite-dimensional space), and the additive translations serve as the "weft" (the intertwining thread that collapses these dimensions into a sequence).

2. The Multiplicative Lattice as an Affine $\mathbb{F}_1$-Space

We begin by isolating the multiplicative structure of the integers, echoing the approach of Connes and Consani in the Arithmetic Site.

Definition 2.1 (k-Smooth Lattices). Let p_k be the k-th prime number. Let S_k denote the set of k-smooth numbers (integers whose prime factors are all $\leq p_k$).

Multiplicatively, S_k is a free commutative monoid generated by $\{p_1, p_2, \ldots, p_k\}$. In the language of absolute algebraic geometry, S_k is exactly the affine space of dimension k over the field with one element, denoted as $\mathbb{A}^k_{\mathbb{F}_1}$.

Over $\mathbb{F}_1$, addition does not exist. The space S_k consists of disconnected points floating in a k-dimensional void. There is no natural metric or ordering between $x, y \in S_k$ unless one is a direct multiple of the other.

3. The Additive-Multiplicative Simplicial Complex ($\mathcal{K}$)

To transition from the static $\mathbb{F}_1$-geometry to the topology of $\mathbb{Z}$, we introduce the "Additive Oracle"—an exogenous probe that attempts to stitch the $\mathbb{A}^k_{\mathbb{F}_1}$ space into a 1-dimensional continuum. We formalize this interaction as a simplicial complex $\mathcal{K}$.

3.1. Simplices and Boundary Operators

- **0-Simplices (C_0):** The vertices are the natural numbers, $x \in \mathbb{N}$.
- **1-Simplices (C_1):** We define two distinct families of directed edges:
  - *Additive Edges (E_A):* α_x denotes the directed edge from x to $x+1$.
  - *Multiplicative Edges (E_M):* $\mu_{x,p}$ denotes the directed edge from x to $x \cdot p$, where p is prime.

The boundary operator $\partial_1 : C_1 \to C_0$ is defined standardly as target minus source:

$$\partial_1(\alpha_x) = (x+1) - (x)$$

$$\partial_1(\mu_{x,p}) = (x \cdot p) - (x)$$

3.2. The Absence of Distributive 2-Simplices

Crucially, while we map 2-simplices for the commutativity of addition and the commutativity of multiplication, **we do not define 2-simplices for the distributive law.** The relation $p(x+1) = px + p$ is treated not as an axiomatic triviality, but as a topological hole (a 1-cycle). This omission is the engine of our topological invariants.

4. Persistent Homology and the Dimensional Filtration

We convert the complex $\mathcal{K}$ into a persistence module by defining a filtration function based on the sequential expansion of the k-smooth lattices.

Definition 4.1 (Filtration Function). Let the discrete time parameter $\epsilon = k$ represent the index of the primes. Let $\mathrm{ind}(x)$ be the index of the largest prime factor of x (with $\mathrm{ind}(1) = 0$). We define the filtration $f : \mathcal{K} \to \mathbb{R}$ as:

- $f(x) = \mathrm{ind}(x)$ for vertices.
- $f(\mu_{x,p_i}) = \max(f(x), i)$ for multiplicative edges.
- $f(\alpha_x) = \max(f(x), f(x+1))$ for additive edges.

This yields a sequence of subcomplexes $\mathcal{K}_0 \subseteq \mathcal{K}_1 \subseteq \cdots \subseteq \mathcal{K}$, where $\mathcal{K}_k$ is the geometric realization of the interaction between the $+1$ oracle and the k-dimensional $\mathbb{F}_1$-lattice.

4.1. H_0 Persistence and Prime Gaps

At time $\epsilon = k$, the multiplicative edges populate instantly, but the additive edges lag severely because adjacent integers are rarely smooth with respect to the same small set of primes (Størmer's Theorem).

Theorem 1 (Prime Gaps as H_0 Death Times). *At filtration step $\mathcal{K}_k$, the space is shattered into disconnected H_0 components ("floating islands"). The "death" of an H_0 component occurs exactly when the lattice expands sufficiently to allow the Additive Oracle to bridge the gap. Consequently, the maximum lifespan of H_0 classes in $\mathcal{K}_k$ encodes the bounds of the prime gaps following p_k.*

5. The Knot Theory of the Arithmetic Braid

Because we omitted the distributive 2-simplices, a path that leaves the origin multiplicatively and returns additively forms a closed H_1 loop. This allows us to map the evaluation of multiplication to formal knot invariants.

5.1. The Distributive Cycle

For any integer x and prime p, evaluating the multiplicative edge $\mu_{x,p}$ using only the Additive Oracle requires traversing $x(p-1)$ distinct α edges. We formally define the **Distributive Cycle** $\mathcal{D}(x,p)$ as:

$$\mathcal{D}(x,p) = \mu_{x,p} - \sum_{i=0}^{x(p-1)-1} \alpha_{x+i}$$

The length (or "weight") of the additive relaxation of this cycle is exactly $xp - x = x(p-1)$.

5.2. The Arithmetic Linking Number $\ell(p,q)$

In a standard symmetric monoidal category, $p \otimes q \cong q \otimes p$. In our computationally braided complex, swapping the order of multiplication induces a topological twist. Consider evaluating 14 via the primes 2 and 7.

- **Path A (7×2):** Jump to 7, then evaluate $\mu_{7,2}$ additively. The additive path requires $(14-7) = \mathbf{7}$ additive steps.
- **Path B (2×7):** Jump to 2, then evaluate $\mu_{2,7}$ additively. The additive path requires $(14-2) = \mathbf{12}$ additive steps.

Definition 5.1 (Computational Linking Number). We define the Arithmetic Linking Number $\ell(p,q)$ between two primes as the absolute difference in the additive weights of their braided evaluations:

$$\ell(p,q) = |p(q-1) - q(p-1)| = |pq - p - pq + q| = |p-q|$$

This invariant proves that multiplication is only computationally symmetric when $p = q$. The

metric distance $|p - q|$ directly measures the "tightness" of the knot created by swapping the prime fibers in the evaluation braid.

5.3. The Path Invariant Theorem (Factorization Topology)

This local knot theory elegantly generalizes to the entire factorization tree of any composite integer N. Let $\Gamma = (p_1, p_2, \ldots, p_n)$ be an ordered sequence of prime factors evaluating to N. The total additive weight $W(\Gamma)$ of evaluating N sequentially is the sum of the additive relaxations of each step. As we sum the total topological weight of the braid, the intermediate terms cancel telescopically:

$$W(\Gamma) = (p_1 p_2 - p_1) + (p_1 p_2 p_3 - p_1 p_2) + \cdots + (N - N/p_n) = N - p_1$$

Theorem 2 (The Fundamental Path Invariant). *The total additive weight of evaluating any composite integer N depends entirely and exclusively on the initial prime dimension entered. For any factorization braid Γ, the total additive topological weight is an invariant given by:*

$$W(\Gamma) = N - p_{base}$$

where p_{base} is the first prime in the operational sequence.

5.4. The Arithmetic Jones Polynomial and Skein Relations

To formally classify these factorization braids, we construct an analogue to the Jones polynomial. Let t be a formal variable where the exponent denotes additive weight.

Definition 5.2 (State Polynomial of a Braid). For a factorization braid $\Gamma = (p_1, p_2, \ldots, p_n)$ resolving to N, with intermediate nodes $N_i = \prod_{j=1}^{i} p_j$, we define the state polynomial $\mathcal{J}_\Gamma(t)$ as the sum of the localized node weights:

$$\mathcal{J}_\Gamma(t) = \sum_{i=1}^{n} t^{N_i}$$

In classical knot theory, Reidemeister moves allow one to continuously deform a knot without changing its topology. In our arithmetic complex, an **Arithmetic Reidemeister Move** consists of swapping adjacent primes p_k, p_{k+1} in the evaluation sequence (where $k > 1$, preserving

p_{base}).

Let Γ_+ and Γ_- be two braids identical everywhere except at an intermediate junction evaluating from a base x. Γ_+ evaluates $x \to xp \to xpq$, while Γ_- evaluates $x \to xq \to xqp$. This swap leaves N_{final} unchanged but alters the internal state polynomial, yielding the **Arithmetic Skein Relation**:

$$\mathcal{J}_{\Gamma_+}(t) - \mathcal{J}_{\Gamma_-}(t) = t^{xp} - t^{xq}$$

Because all intermediate nodes telescope out in the final boundary evaluation, the global knot invariant of the equivalence class of braids (all factorizations of N starting with p_{base}) collapses to a strictly conserved monomial:

$$\mathcal{V}_{N,p_{base}}(t) = t^{N-p_{base}}$$

This proves that internal prime permutations are homology-preserving transformations; they shift intermediate tension without altering the global topological mass of the integer.

6. The $\mathbb{F}_1$ Tangent Space and Canonical Roots

The telescopic cancellation yielding $W(\Gamma) = N - p_{base}$ is not merely an algebraic curiosity; it is the manifestation of a conservation law rooted deeply in the differential geometry of the Arithmetic Topos over $\mathbb{F}_1$.

6.1. The Origin and Tangent Vectors of $\mathbb{F}_1$

In absolute algebraic geometry, the field with one element $\mathbb{F}_1$ lacks addition. The "origin" of an affine space $\mathbb{A}^k_{\mathbb{F}_1}$ is the multiplicative identity, 1.

To build an integer additively, the Additive Oracle must step away from the origin. In differential geometry, stepping away from an origin requires the selection of a **tangent vector**. In our lattice, the prime numbers $\{p_1, p_2, \dots\}$ serve exactly as the basis vectors of the tangent space at the origin 1, denoted $T_1(\mathbb{A}^\infty_{\mathbb{F}_1})$.

6.2. The Conservation of Topological Cost

When an evaluation braid Γ selects p_{base} as its first operation, it is choosing a canonical tangent vector—a "principal section"—to enter the multiplicative lattice.

Once p_{base} is selected, the rest of the factorization unfolds entirely within the "light cone" (or tangent bundle) of that vector. Because multiplicative operators commute dimensionally, all subsequent operations are effectively **isometries** within that specific tangent bundle.

Theorem 3 (Conservation of Arithmetic Topology). *The topological cost of factorization is conserved within the* $\mathbb{F}_1$ *tangent space. The distance from the target* N *to the canonical root* p_{base} *is invariant under internal permutations because all such paths are homotopic within the tangent bundle* $T_{p_{base}}(\mathcal{K})$.

Consequently, to evaluate N starting from a *different* base prime q_{base} requires exiting the current tangent bundle, passing back through the singular origin at 1, and re-entering the lattice along an orthogonal axis. This is why changing p_{base} alters the fundamental topological class (the weight $N - p_{base}$) of the integer knot.

7. The Dimensional Expansion Theorem

We can now articulate the central thesis of this framework. The emergence of prime numbers is not a random arithmetic sequence, but a forced geometric base change.

As the Additive Oracle probes the k-dimensional $\mathbb{A}^k_{\mathbb{F}_1}$ lattice, it inevitably encounters unresolvable H_1 tensions (distributive truths that cannot be proven additively because the required vertices do not exist in S_k).

The Additive Oracle acts as the functor of base change $\otimes_{\mathbb{F}_1}\mathbb{Z}$. To preserve the continuity of the 1-dimensional additive continuum, the space is physically forced to undergo a dimensional expansion. The next prime p_{k+1} is the minimal topological patch—a new orthogonal axis in the $\mathbb{F}_1$ tangent space—injected into the lattice to close the broken H_1 cycles and merge the shattered H_0 islands.

8. Conclusion and Future Directions

By treating the primes as dimensional lattice generators and the $+1$ operation as an exogenous probing oracle, we successfully map the integers into a persistent homology framework. This model unifies computational complexity, the arithmetic site over $\mathbb{F}_1$, and the distribution of prime numbers into a cohesive topological space.

The discovery of the Arithmetic Linking Number, the Arithmetic Skein Relation, and the tangent space conservation law ($W(\Gamma) = N - p_{base}$) opens immediate avenues for future research. In particular, applying the Arithmetic Jones Polynomial to semi-primes offers a strictly geometric framework for modeling RSA cryptography, where prime factorization is rendered equivalent to computing the topological unknotting number of a braid across orthogonal $\mathbb{F}_1$ tangent spaces.